

BOSONIZE-LEAN: Phase 2 Formalization Blueprint

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1 Phase 2 Scope Constraints

Based on the Phase 2 directives and critiques, this phase is strictly limited to the formulation of the Bosonization Dictionary's density sector. Klein factors, Mattis-Mandelstam, and Luttinger interaction are deferred to subsequent phases. This document exactly mirrors the required Lean 4 stubs in `Bosonize/Stub/Phase2_Stubs.lean`.

2 2.1 Density Operators

We define the density modes as the Fourier transform of the normal-ordered fermion density. Normal ordering subtracts the macroscopic vacuum expectation value to keep the operator finite on the budget subspace.

```
section Densities
```

```
-- The raw density operator  $\rho(m) = \sum c^\dagger_{q+m} c_q$  -/
```

```
def density_mode (m : LambdaDual L) : Module.End Complex  
  (FockSpace (Chirality x LambdaDual L)) :=
```

```
-- Note to Blue Team: implement using Finset.sum over  
  q in LambdaDual
```

```
  sorry
```

```
-- Vacuum expectation value of the density mode -/
```

```
def density_vev (m : LambdaDual L) : Complex :=
```

```

-- Tr( |0><0| \rho(m) )
sorry

/-- Normal ordered density mode :\rho(m): -/
def no_density_mode (m : LambdaDual L) : Module.End
  Complex (FockSpace (Chirality x LambdaDual L)) :=
    density_mode m - LinearMap.smulRight (LinearMap.id) (
      density_vev m)

/-- The zero mode of the normal ordered density is
    exactly the particle number shift -/
lemma no_density_zero_eq_N :
  no_density_mode 0 = N_op := sorry

end Densities

```

3 2.2 Heisenberg Algebra on the Budget Subspace

The Schwinger term arises algebraically. The commutator is evaluated strictly on the finite Energy Budget Subspace $\mathcal{B}_{K,N_{max}}$ to rigorously avoid infinite limits and domain issues.

`section` HeisenbergCCR

```

/--
  The Exact Schwinger Term on the Budget Subspace.
  Requires the momentum modes m, n to be small enough
  such that they do not
  excite states out of the finite band L when acting on
  a state in the budget K.
-/
lemma comm_no_density_modes {K Nmax : Nat} {m n :
  LambdaDual L}
  (h_band : (m.val : Int) + (n.val : Int) + K < L / 2) :
  \forall (psi : BudgetSpace L K Nmax),
    (no_density_mode m * no_density_mode n -
      no_density_mode n * no_density_mode m) \psi.val =

```

```

      (if m.val + n.val = 0 then (m.val : Complex) else 0)
      \bullet \psi.val := sorry

end HeisenbergCCR

```

4 2.3 Sugawara Construction

The construction of the bosonic kinetic energy operator. On the budget subspace, this is rigorously proven equivalent to the free linearized fermionic Hamiltonian.

```

section Sugawara

/-- The bosonic kinetic energy operator -/
def sugawara_op : Module.End Complex (FockSpace (
  Chirality x LambdaDual L)) :=
  \sum m in (Finset.filter (fun x : LambdaDual L => (x.
    val : Int) > 0) Finset.univ),
    no_density_mode (-m) * no_density_mode m

/--
  The free fermion Hamiltonian equals the Sugawara
  bosonic Hamiltonian
  on the budget subspace, up to the zero-mode energy
  shift.
-/
lemma sugawara_equivalence {K Nmax : Nat} (h_band : K <
  L / 4) :
  \forallall (\psi : BudgetSpace L K Nmax),
    -- H_0 \psi = (H_sug + N(N+1)/2) \psi
    sorry

end Sugawara

```

5 2.4 Haldane Completeness

Proving that the set of states generated by acting with arbitrary polynomials of $\rho(-m)$ forms a complete basis for the budget subspace via integer partition

counts.

`section` Completeness

`--`

Dimension count showing that the number of linearly
independent states
generated `by` $\rho(-m)$ matches the dimension of the
fixed- N budget subspace,
proving that the bosons completely span the fermionic
degrees of freedom.

`-/`

`lemma` haldane_completeness $\{K\ N : \text{Nat}\}$ (h_band : $K < L /$
4) :
Module.finrank Complex (`sorry` : Submodule Complex (
FockSpace (Chirality x LambdaDual L))) =
Nat.Partition.partitions $K := \text{sorry}$

`end` Completeness